

Problem

The highest-frequency component of a voice signal is 3.4 kHz. What is the Nyquist rate of the sampler of the voice signal?

Step-by-step solution

Step 1 of 1

The highest frequency component of voice signal = 3.4 KHz

We know that,

$$N_s \geq 2f_m \quad [f_m = \text{Highest Modulating frequency}]$$

$$\text{or,} \quad N_s \geq 2 \times 3.4 \text{ KHz}$$

$$\text{or,} \quad N_s \geq 6.8 \text{ KHz}$$

$$\text{or,} \quad \text{Nyquist rate of the sampler of the voice signal is } \boxed{6.8 \text{ KHz}}$$